

A formal and reproducible framework for auditing observable behavior in qualitative biological network models

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Qualitative biological networks are often compared structurally, although similar topology does not ensure that executable models preserve event order, branching, or hidden steps. We present a formal, reproducible framework that maps Petri-net and logical models to finite reachable labeled transition systems under a declared observable interface. An implication-aware classifier reports strong bisimilarity, weak bisimilarity, mutual or one-way weak simulation, or non-comparability; structural and finite-trace scores remain separate diagnostics. We prove greatest-fixed-point correctness and termination for the implemented relation-deletion algorithm and characterize the relation hierarchy, controlled silent refinement, and trace-equivalent branching. In six construction-level cases, weak bisimulation classified 6/6 correctly, compared with 5/6 for truncated traces, 4/6 for LTS-GDA, 3/6 for a structural profile, and 2/6 for PN-GDDA. Python and mCRL2 agreed on 42 strong/weak decisions, and an independent simulation game agreed on all 67,600 ordered comparisons in the declared small-system universe. Public GINsim models, HPN-DREAM/CASPOTS families, and curated Petri nets test cross-format execution, held-out concordance, update semantics, and interface dependence. The framework audits explicit models; it does not establish predictive performance, cell-line specificity, or organism-level equivalence.

Keywords: Qualitative biological networks; Petri nets; labeled transition systems; weak bisimulation; model checking; Boolean networks.

1. Introduction

Qualitative biological networks represent regulatory systems, signaling pathways, Boolean rules, and Petri-net mechanisms when kinetic information is limited. Models intended to describe related processes may nevertheless originate from different organisms, inference pipelines, formalisms, or update semantics. Their comparison is therefore a recurring computational biology problem, not only a graph-matching task. Petri nets provide explicit causality, conflict, and concurrency, while logical models make discrete regulatory assumptions executable.¹⁻⁴

Structural similarity does not determine observable executable behavior. Graphlet methods sensitively compare local motifs; PN-GDDA, for example, uses

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151 directed bipartite Petri-net graphlets and 592 published orbit slots.⁵ By design, such a score does not encode transition labels, initial marking, or recursive reachable branching. Conversely, internal refinement can alter topology while preserving the behavior exposed at a chosen interface. Structure and behavior are therefore complementary evidence layers.

Linear traces add event order but omit where alternatives become committed. Two models can generate the same finite sequences while resolving a biological choice at different states. Branching-time equivalences retain that information, and simulation preorders report asymmetric containment when every behavior of one model can be matched by another. Recent formal work on asynchronous Boolean networks likewise highlights the role of reachability and branching structure in phenotypic analysis.⁶

Let N_1 and N_2 be executable labeled models and let Σ be a declared observable interface. We formulate the comparison as

$$(N_1, N_2, \Sigma) \mapsto (\mathcal{L}(N_1), \mathcal{L}(N_2)) \mapsto C_\Sigma(\mathcal{L}(N_1), \mathcal{L}(N_2)), \quad (1.1)$$

where $\mathcal{L}(N_i)$ is the finite reachable labeled transition system and C_Σ returns the strongest supported class among strong bisimilarity, weak bisimilarity, mutual weak simulation, either one-way weak simulation, and non-comparability. We call the reporting priority *graded*: a stronger relation suppresses weaker implied labels. This is an implication-aware classifier, not an algebraic grading. Structural and truncated-trace measures remain attached diagnostics rather than substitutes for C_Σ .

The formal relations are standard, but their use here is made auditable. We define the observable interface and simulation orientation, prove greatest-fixed-point correctness for the executed relation-deletion algorithm, and establish the hierarchy supporting the classifier. Controlled silent refinement and minimal counterexamples separate structure, traces, and branching. The implementation is checked against construction-level ground truth, mCRL2, and an independently coded attacker-defender game that is exhaustive within the declared universe of one- and two-state systems over $\{a, \tau\}$.

Biological evaluation then exercises different sources of heterogeneity. Public GINsim models test SBML-qual/GINML import and asynchronous versus synchronous updates.^{7–10} HPN-DREAM/CASPOTS families connect independently inferred Boolean rules to held-out perturbation-response data.^{11,12} Curated plant and animal Petri nets expose dependence on the observational interface. These cases evaluate cross-format use and empirical concordance; they are not premises of the mathematical results.

The contributions are:

- (i) a formal bioinformatics framework that audits two executable qualitative biological models under a declared interface;
- (ii) mathematical guarantees for the implemented classifier, together with properties and counterexamples distinguishing structure, traces, and branching;

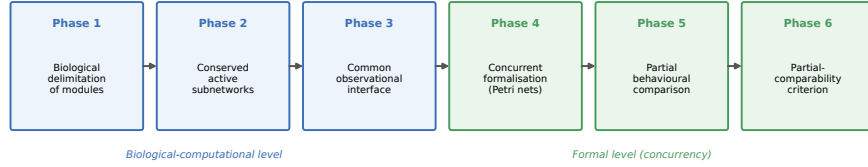


Fig. 1. Six-stage workflow. Stages 1–3 delimit the model-comparison question and observational interface; stages 4–6 construct executable Petri nets, compare their reachable behavior, and assign a model-level relation.

- (iii) independent computational validation through six controlled cases, mCRL2, 67,600 attacker–defender comparisons, structural/trace baselines, and observed computational scaling; and
- (iv) applications to public GINsim models, HPN-DREAM/CASPOTS families, and curated Petri nets, supported by public code, data provenance, tests, and an executed notebook.

The comparison layers retain different information. PN-GDDA and the structural profile summarize topology; traces retain order and silent abstraction but not branching; LTS-GDA retains labeled local motifs. The integrated audit additionally matches recursive branching and directional containment, while held-out data remain a separate evidence layer.

The operational sequence is summarized in Fig. 1. Biological delimitation and interface declaration precede formal encoding, reachability construction, and classification. This separation makes the modeling assumptions visible before any equivalence calculation is performed.

2. Formal framework for observable biological-model comparison

2.1. Labeled Petri nets and induced transition systems

Definition 1 (Labeled place/transition net). A labeled place/transition net is a tuple

$$N = (P, T, \text{pre}, \text{post}, \ell, m_0), \quad (2.1)$$

where P and T are finite sets of places and transitions, $\text{pre}, \text{post} : T \rightarrow \mathbb{N}^P$ are the input and output multisets, $m_0 \in \mathbb{N}^P$ is the initial marking, and $\ell : T \rightarrow \Sigma \cup \{\tau\}$ assigns either an observable action in the declared interface Σ or the internal action τ .

Under standard interleaving semantics, t is enabled at m when $\text{pre}(t) \leq m$ componentwise; firing it changes m to $m - \text{pre}(t) + \text{post}(t)$. The implementation uses a worklist to enumerate reachable markings explicitly. A state cap and a token

bound make failure to establish a finite state space explicit rather than silently truncating the relation calculation. All curated Petri nets used in Section 4 are safe, but the definitions below apply to every finite reachable system.

Definition 2 (Finite labeled transition system). A finite labeled transition system (LTS) is

$$L = (S, \Sigma_\tau, \longrightarrow, s_0), \quad \Sigma_\tau = \Sigma \cup \{\tau\}, \quad (2.2)$$

where S is a finite state set, $s_0 \in S$, and $\longrightarrow \subseteq S \times \Sigma_\tau \times S$. We write $s \xrightarrow{a} s'$ for $(s, a, s') \in \longrightarrow$. The reachable semantics of the net in Definition 1 is denoted by $\mathcal{L}(N)$.

2.2. Observable and weak transitions

Definition 3 (Silent closure and weak target sets). For $s \in S$, let

$$\text{Cl}_\tau(s) = \{u \in S : s \xrightarrow{\tau^*} u\}. \quad (2.3)$$

For $a \in \Sigma$, define

$$W_a(s) = \{v : \exists u, w \quad s \xrightarrow{\tau^*} u \xrightarrow{a} w \xrightarrow{\tau^*} v\}, \quad W_\tau(s) = \text{Cl}_\tau(s). \quad (2.4)$$

We write $s \xRightarrow{a} v$ when $v \in W_a(s)$.

The convention $W_\tau = \text{Cl}_\tau$ is important: a direct silent move on one side may be matched by zero or more silent moves on the other. It is also the convention implemented by the Python engine and used in the AUT exports checked by mCRL2.

2.3. Simulation and bisimulation

Let $L_i = (S_i, \Sigma_\tau, \longrightarrow_i, s_i^0)$ for $i \in \{1, 2\}$.

Definition 4 (Weak simulation). A relation $R \subseteq S_1 \times S_2$ is a weak simulation from L_1 to L_2 if, whenever $(x, y) \in R$ and $x \xrightarrow{a}_1 x'$, there is $y' \in W_a^2(y)$ such that $(x', y') \in R$. We write

$$L_1 \preceq_w L_2 \quad (2.5)$$

if (s_1^0, s_2^0) belongs to some such relation. Thus the right-hand system L_2 simulates the left-hand system L_1 ; the left-hand behavior is contained in the right-hand behavior under the interface.

Definition 5 (Weak and strong bisimulation). A relation $R \subseteq S_1 \times S_2$ is a weak bisimulation if it satisfies the condition in Definition 4 and, for every $(x, y) \in R$ and $y \xrightarrow{a}_2 y'$, there is $x' \in W_a^1(x)$ with $(x', y') \in R$. The initial systems are weakly bisimilar, written $L_1 \sim_w L_2$, when their initial pair belongs to such a relation. Strong bisimulation, written $L_1 \sim_s L_2$, replaces every weak target set $W_a^i(x)$ by the direct target set $\{x' : x \xrightarrow{a}_i x'\}$, including for $a = \tau$.

These are standard branching-time relations.^{13,14} The explicit orientation in (2.5) prevents the common ambiguity between “simulates” and “is simulated by” in one-way results.

2.4. Weak trace language

Definition 6 (Exact and truncated weak traces). The finite weak observable language of L is

$$L_w(L) = \{a_1 \cdots a_r \in \Sigma^* : s_0 \xRightarrow{a_1} \cdots \xRightarrow{a_r} s \text{ for some } s \in S\}. \quad (2.6)$$

Its depth- k restriction is $L_{\leq k}(L) = \{u \in L_w(L) : |u| \leq k\}$. The diagnostic distance is

$$d_k(L_1, L_2) = 1 - \frac{|L_{\leq k}(L_1) \cap L_{\leq k}(L_2)|}{|L_{\leq k}(L_1) \cup L_{\leq k}(L_2)|}. \quad (2.7)$$

The implementation computes the prefix-closed truncated sets in Definition 6. The statement $d_k = 0$ is therefore not identified with exact weak-trace equivalence. Exact weak-trace results in this study come only from mCRL2 or from a finite construction for which the complete language is enumerated analytically.

2.5. Graded behavioral classification

Definition 7 (Graded classifier). For finite L_1, L_2 over the declared interface Σ , define $C_\Sigma(L_1, L_2)$ by the following priority:

$$C_\Sigma(L_1, L_2) = \begin{cases} \text{strong,} & L_1 \sim_s L_2, \\ \text{weak,} & L_1 \not\sim_s L_2 \text{ and } L_1 \sim_w L_2, \\ \text{mutual,} & L_1 \not\sim_w L_2, L_1 \preceq_w L_2, L_2 \preceq_w L_1, \\ \text{left_in_right,} & L_1 \preceq_w L_2, L_2 \not\preceq_w L_1, \\ \text{right_in_left,} & L_2 \preceq_w L_1, L_1 \not\preceq_w L_2, \\ \text{noncomparable,} & \text{otherwise.} \end{cases} \quad (2.8)$$

The output is a reporting class, not a newly proposed equivalence relation.

Proposition 1 (Exclusive reporting). *The classifier in Definition 7 assigns exactly one class to every ordered pair of finite LTSs.*

Proof. The first two cases are selected before any simulation-only case. If neither is selected, the two simulation predicates have exactly four Boolean combinations: both true, only the first true, only the second true, or both false. These are the last four cases of (2.8), respectively. The cases are exhaustive and disjoint by their stated guards. $\square$

2.6. Fixed-point computation and formal properties

For $a \in \Sigma_\tau$, let $D_a^i(x) = \{x' : x \xrightarrow{a}_i x'\}$ and $U = S_1 \times S_2$. Define operators on $\mathcal{P}(U)$ by

$$\Phi_{\preceq}(R) = \{(x, y) \in U : \forall a, x' \in D_a^1(x) \exists y' \in W_a^2(y), (x', y') \in R\}, \quad (2.9)$$

$$\Phi_w(R) = \{(x, y) \in \Phi_{\preceq}(R) : \forall a, y' \in D_a^2(y) \exists x' \in W_a^1(x), (x', y') \in R\}, \quad (2.10)$$

$$\begin{aligned} \Phi_s(R) = \{(x, y) \in U : \forall a, x' \in D_a^1(x) \exists y' \in D_a^2(y), (x', y') \in R, \\ \forall a, y' \in D_a^2(y) \exists x' \in D_a^1(x), (x', y') \in R\}. \end{aligned} \quad (2.11)$$

Theorem 1 (Correctness of relation deletion). *For each $\Phi \in \{\Phi_{\preceq}, \Phi_w, \Phi_s\}$, the operator is monotone. The implemented algorithm starts with $R = U$, repeatedly deletes any currently examined pair $z \notin \Phi(R)$, and stops after a complete pass with no deletion. On finite LTSs it terminates and returns the greatest fixed point $\nu\Phi$. Consequently, testing membership of (s_1^0, s_2^0) computes, respectively, weak simulation in the orientation of Definition 4, weak bisimilarity, and strong bisimilarity.*

Proof. If $R \subseteq Q$, every witness pair in R remains available in Q ; hence $\Phi(R) \subseteq \Phi(Q)$ for each operator. Every deletion strictly decreases the finite relation, so termination requires at most $|U|$ deletions.

Let $G = \nu\Phi$. The invariant $G \subseteq R$ holds initially. If it holds before a deletion and $z \in G$, then $z \in \Phi(G) \subseteq \Phi(R)$, so z cannot be deleted. Thus G remains a subset of the terminal relation R_* . A final pass without deletion gives $R_* \subseteq \Phi(R_*)$. Conversely, each $z \notin R_*$ was deleted from some R_z with $z \notin \Phi(R_z)$. Since $R_* \subseteq R_z$, monotonicity implies $\Phi(R_*) \subseteq \Phi(R_z)$ and therefore $z \notin \Phi(R_*)$. Hence $R_* = \Phi(R_*)$. Because R_* is a fixed point, $R_* \subseteq G$; together with the invariant this yields $R_* = G$. The three operators are exactly the transfer clauses of Definitions 4 and 5. $\square$

Remark 1 (Resource bound). Writing $n_i = |S_i|$, the candidate relation stores $O(n_1 n_2)$ pairs and permits at most $n_1 n_2$ deletions and $n_1 n_2 + 1$ passes. Each pass scans current pairs, outgoing moves, and matching direct or weak targets. Silent closures are cached; visible weak targets are constructed on demand. We therefore report these parameters instead of claiming an unjustified tight time bound.

Proposition 2 (Hierarchy of behavioral relations). *For finite LTSs under Definitions 3–6,*

$$L_1 \sim_s L_2 \implies L_1 \sim_w L_2 \implies (L_1 \preceq_w L_2 \wedge L_2 \preceq_w L_1) \implies L_w(L_1) = L_w(L_2). \quad (2.12)$$

The converses do not hold in general.

Proof. A direct match is a permitted weak match, and each direction of a weak bisimulation is a weak simulation. If $L_1 \preceq_w L_2$, induction along any finite concrete path matches observable moves by $\tau^* a \tau^*$ and silent moves by τ^* , preserving related successors. Erasing τ gives $L_w(L_1) \subseteq L_w(L_2)$; the reverse simulation gives equality.

Proposition 3 and Example 1 refute the first and last converses. For the middle converse, compare $a.(b + c) + a.b$ with $a.(b + c)$: each simulates the other, but the $a.b$ successor cannot be bisimilar to a state offering both b and c . $\square$

Proposition 3 (Controlled silent edge refinement). *Let L contain $p \xrightarrow{a} q$ with $a \in \Sigma$ and $p \neq q$. Construct L' by copying all original states and transitions except this edge, and replacing it by*

$$\bar{p} \xrightarrow{a} r \xrightarrow{\tau} \bar{q}, \quad (2.13)$$

where r is fresh and has no other incident behavior. Then $L \sim_w L'$. Repeated refinements of this form also preserve weak bisimilarity. If p and $\bar{p}$ are initial, the selected edge is the only a -edge from $\bar{p}$ in L' , and every a -successor of p in L has no outgoing τ transition, then $L \not\sim_s L'$.

Proof. Let $R = \{(x, \bar{x}) : x \in S\} \cup \{(q, r)\}$. For every copied pair other than the source, direct moves are copied. At $(p, \bar{p})$, the original edge is matched by $\bar{p} \xrightarrow{a} r \xrightarrow{\tau} \bar{q}$, while $\bar{p} \xrightarrow{a} r$ is matched by $p \xrightarrow{a} q$ and ends in (q, r) . At (q, r) , moves from q are matched after $r \xrightarrow{\tau} \bar{q}$; that silent move is itself matched by zero silent moves at q . Hence R is a weak bisimulation, and repetition follows by transitivity. Under the additional conditions, a strong match for $\bar{p} \xrightarrow{a} r$ must relate r to an a -successor x of p , but $r \xrightarrow{\tau} \bar{q}$ would require the excluded direct τ -move from x . Thus strong bisimilarity fails. $\square$

Proposition 4 (Limit of label-blind structural comparison). *Let M be any comparator determined only by unlabeled structure and invariant under relabeling of transitions. Then M cannot characterize weak-trace equality, weak simulation or weak bisimilarity for labeled systems in general.*

Proof. Take two systems with the same two states and one directed edge. Label the edge a in L_a and $b \neq a$ in L_b . Their unlabeled structures are identical, so $M(L_a, L_b) = M(L_a, L_a)$. Yet their languages are $\{\epsilon, a\}$ and $\{\epsilon, b\}$, and neither observable move can be matched by the other system. Thus no decision based only on M can characterize the stated labeled relations. $\square$

Proposition 4 does not identify a defect in PN-GDDA. It states that a label-blind structural method answers a different question and cannot substitute for a labeled behavioral relation.

Example 1 (Equal traces, different branching). Let L_{late} have transitions

$$\ell_0 \xrightarrow{a} \ell_1, \quad \ell_1 \xrightarrow{b} \ell_b, \quad \ell_1 \xrightarrow{c} \ell_c, \quad (2.14)$$

and let L_{early} have

$$e_0 \xrightarrow{a} e_b, \quad e_0 \xrightarrow{a} e_c, \quad e_b \xrightarrow{b} e'_b, \quad e_c \xrightarrow{c} e'_c. \quad (2.15)$$

Both exact prefix-closed languages are $\{\epsilon, a, ab, ac\}$. Nevertheless, $L_{\text{late}} \not\sim_w L_{\text{early}}$. There are no silent moves, and matching either early a -successor with ℓ_1 leaves one of the two late actions unmatched. Directionally, $L_{\text{early}} \preceq_w L_{\text{late}}$: the late-choice state can match the selected b or c continuation of either early branch. The reverse $L_{\text{late}} \preceq_w L_{\text{early}}$ fails because no single early a -successor offers both continuations. Thus exact trace equality does not determine branching-time equivalence or simulation orientation.

2.7. Computational implementation

Algorithm 1 summarizes the executable path. Reachability and the formal relations are exact whenever the finite state space is completed below the declared state and token caps. The distance d_k , PN-GDDA, LTS-GDA, and the structural profile are diagnostics rather than formal relations.

Algorithm 1 (Executable comparison pipeline).

- (i) Enumerate the reachable LTS from the initial marking or logical state by a worklist; stop with an explicit error if a cap is exceeded.
- (ii) Cache $\text{Cl}_\tau(s)$ and construct $W_a(s)$ by collecting a -successors from the closure and the closures of their targets.
- (iii) Initialize $R = S_1 \times S_2$ and delete a pair immediately when it fails the selected transfer operator; repeat complete passes until none is deleted.
- (iv) Evaluate strong and weak bisimilarity and both simulation directions, apply Eq. (2.8), and attach trace and structural diagnostics.

3. Computational implementation and validation

3.1. Construction-level benchmark and comparator study

Six deterministic pairs were specified before biological applications. A 12-step labeled workflow supplies identity, controlled silent subdivision, label mismatch, label-order swap, and an added observable branch. The sixth pair is $a.(b+c)$ versus $a.b+a.c$ from Example 1. Table 1 gives construction-level and computed classes.

Weak bisimulation recovered all six binary weak-equivalence decisions. Truncated trace equality recovered 5/6 but accepted the trace-equivalent branching pair; LTS-GDA recovered 4/6, the structural profile 3/6, and direct PN-GDDA 2/6. PN-GDDA assigned 1.0 to identity, silent refinement, label mismatch, and order swap, while added-branch and branching pairs scored 0.9676 and 0.9657. No threshold over the six PN-GDDA scores exceeded 4/6 because two negative controls tied the positive controls at 1.0. Figure 2 reports all predeclared decisions.

The structural profile averages state-count, edge-count, label-multiset, and out-degree-multiset similarities. The trace diagnostic declares a depth-8 match when $d_8 = 0$. PN-GDDA was implemented from its published definition using 151 directed

Table 1. Controlled benchmark; the expected class is not used by the classifier.

Case	Change	Expected	Observed
Identity	exact copy	strong equivalence	strong equivalence
Silent refinement	insert internal steps	weak equivalence	weak equivalence
Label mismatch	replace an observable	not comparable	not comparable
Order swap	preserve counts, change order	not comparable	not comparable
Added branch	add an outcome	reference in candidate	reference in candidate
Early/late choice	move decision point	candidate in reference	candidate in reference

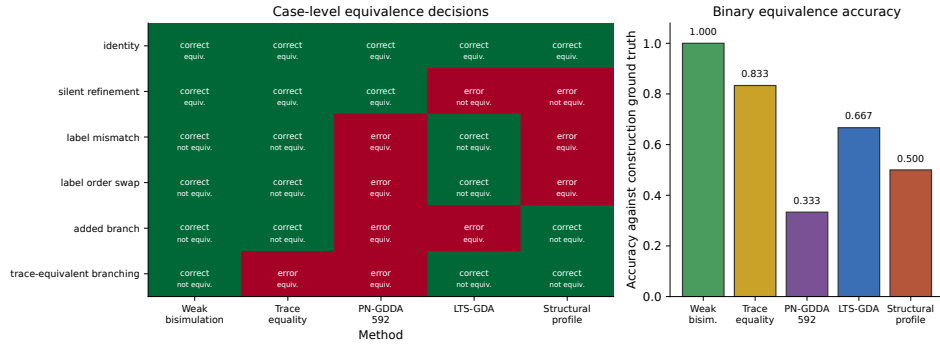


Fig. 2. Controlled benchmark. Left: binary decision by case and method. Right: accuracy against labels fixed by construction. Trace equality loses branching; PN-GDDA is label-blind; LTS-GDA retains local labels but does not abstract silent refinement or recursively match successors.

bipartite graphlets and the 592 published/Holmes orbit slots.⁵ For orbit j ,

$$N_G^j(k) = \frac{D_G^j(k)/k}{\sum_{r>0} D_G^j(r)/r}, \quad A^j(G, H) = 1 - \frac{\left[\sum_k (N_G^j(k) - N_H^j(k))^2 \right]^{1/2}}{\sqrt{2}}, \quad (3.1)$$

and PN-GDDA is the mean A^j . Synthetic LTSs use an exact state-machine Petri net encoding; curated pairs use their native nets. The 0.9 threshold is a fixed diagnostic decision, not an equivalence relation. LTS-GDA instead combines four reachable-graph orbits with labeled edge, two-step, divergence, and convergence motifs.

The catalog audit recovered 151 graphlets and 592 slots from hash-verified Holmes 1.1.1/2.0.1.2 JARs. A direction- and type-preserving automorphism partition has 576 distinct orbits; using it is reported only as sensitivity. A Java probe matched every Holmes 1.1.1 topology/root assignment. On a four-node test net and its one-arc deletion, Python and Holmes scores were 0.9872027465870733 and

0.987202746587073, agreeing to 12 decimal places. Holmes is an independent software reference for the structural score.¹⁵

3.2. *Independent formal oracles*

Each analyzed LTS was exported in Aldebaran AUT format and evaluated with `ltscompare` from hash-pinned mCRL2 202607.0.¹⁶ The tool independently checked strong bisimilarity, weak bisimilarity, and exact weak-trace equivalence with `tau` hidden. Its used interface does not expose the weak simulation preorder, so no one-way simulation is attributed to mCRL2.

A separately coded attacker-defender game grows the least set of positions from which an unmatched move can be forced. It agreed with the production simulation algorithm on all 67,600 ordered pairs among all 260 one- and two-state LTSs over $\{a, \tau\}$. This is exhaustive within that finite universe, not a proof for arbitrary systems. Across the six synthetic pairs, six public-model controls, and nine HPN comparisons below, Python and mCRL2 agreed on all 42 reported strong/weak decisions. Theorem 1, finite exhaustive testing, and external-tool comparison are distinct evidence layers.

3.3. *Observed computational scaling and reproducibility*

Linear workflows of 8, 16, 32, 64, 96, and 128 transitions were compared with an identity and a silently refined copy. Median runtime over three repetitions rose from about 0.4 ms for nine states to about 0.6 s for the largest pairs. The 128-step identity relation had 16,641 candidate pairs; refinement increased it to 18,705 pairs over 129 and 145 states. Every class remained as constructed (the complete plot and table are archived in the repository). Timings used Python 3.11.8 on an Apple M3 Pro with 18 GiB memory and are machine-dependent.

The Python standard library implements reachability, formal relations, parsers, and controlled models. NumPy, pandas, Matplotlib, NetworkX, pydot, and Graphviz support tabulation and figures. Fixed seeds, source hashes, machine-readable outputs, unit tests, an executed notebook, environment files, and continuous integration are public. Runtime CSV values are excluded from byte-level regression because they depend on hardware; all scientific outcomes and other result tables are deterministic.

4. *Biological network applications*

4.1. *Cross-format validation with public GINsim models*

Two externally authored GINsim models test the complete import-to-comparison path: the 13-component mammalian cell cycle in SBML-qual and the four-component multilevel p53-Mdm2 model in GINML.^{7–10} Files, URLs, and SHA-256 values are archived. Unitary-asynchronous updates move one enabled component by

Table 2. Public GINsim models and control outcomes. S/W/T denote strong, weak, and exact weak-trace equivalence.

Model	Format	States	Edges	Copy	Silent	Perturbed
Mammalian cell cycle	SBML-q	826	3,413	Y/Y/Y	N/Y/Y	N/N/N
p53–Mdm2 damage	GINML	17	26	Y/Y/Y	N/Y/Y	N/N/N

Table 3. HPN-DREAM families and structure-only representatives.

Cell line	Family	Medoid row ^a	Clauses	Mean Jaccard
BT20	72	24	29	0.815
BT549	191	137	25	0.854
MCF7	21	9	23	0.851

Note: ^aZero-based row; selected from model structure only.

one level and label its direction. Each base model was compared with an exact copy, five silent subdivisions, and a reachable-label perturbation. All six controls recovered their predeclared strong/weak/exact-trace outcomes in Python and mCRL2 (Table 2).

Synchronous execution supplied a semantic stress test. It reached 13 states for the cell-cycle model, all among 826 asynchronous states (Jaccard 0.0157), and eight states for p53–Mdm2, all among 17 asynchronous states (Jaccard 0.471). Thus the controls validate parsing, execution, export, and comparison across formats while demonstrating that reachability is update-semantics dependent.

4.2. HPN-DREAM/CASPOTS held-out analysis

The public HPN-DREAM phosphoproteomic series and CASPOTS Boolean families contain 72 BT20, 191 BT549, and 21 MCF7 networks.^{11,12} One representative per cell line was selected before test responses were read: the model with maximum mean within-family Jaccard similarity between active clauses, with exact lexicographic tie-breaking. This structure-only rule selected zero-based rows 24, 137, and 9 (Table 3); an anti-leakage test forbids access to test files during selection.

The interface is the intersection of seven measured PI3K/MAPK/mTOR readouts. The three common test perturbations are mTOR inhibition, IGF1 plus mTOR inhibition, and 4EBP1 plus mTOR inhibition. Measured time-zero values initialize readouts; stimuli and inhibitors are clamped; other relevant nodes start at zero. Primary asynchronous LTSs label interface updates and hide all others. The nine cross-cell pair-condition comparisons were repeated synchronously; this stress test is not a most-permissive analysis. CASPOTS itself evaluates asynchronous trajectories.^{12,17}

Experimental distance is RMSE over common post-zero times and readouts.

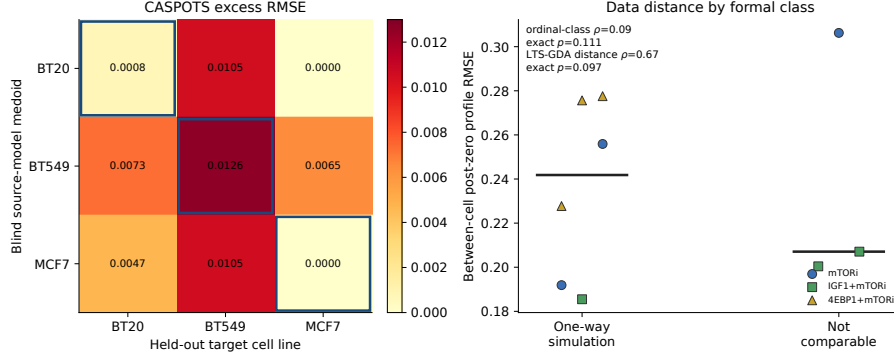


Fig. 3. Held-out HPN-DREAM analysis. (a) CASPOTS excess RMSE by source medoid and target; boxes mark native pairs. (b) Between-cell profile RMSE by formal class. Exact tests are condition-stratified; neither panel establishes native cell-line specificity.

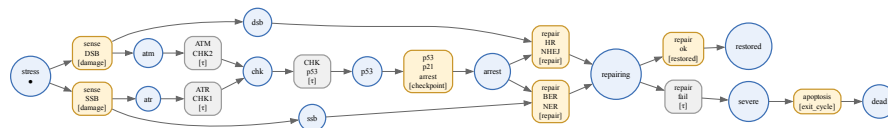
Spearman associations use an exact one-sided permutation of the three cell-pair labels within each condition ($3!^3 = 216$). CASPOTS/Clingo compatibility scores consume optimization to completion, use deterministic secondary tie-breaks, and were reproduced twice. Native specificity uses the exact sign permutation of three native-minus-mean-cross differences.

Selected LTSs had 2–44 states. None of the nine cross-cell pairs was weakly bisimilar: six had one-way simulation and three were not comparable. The ordinal class association with experimental distance was $\rho = 0.091$ ($p = 0.111$); truncated trace distance gave $\rho = -0.596$ ($p = 0.556$). LTS-GDA had the largest observed association, $\rho = 0.669$, but its exact $p = 0.097$ was inconclusive. Synchronous updating preserved seven of nine classes.

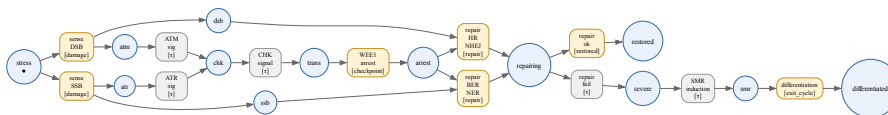
Native medoid RMSE was 0.33024 (BT20), 0.31335 (BT549), and 0.27726 (MCF7), against discretization RMSE 0.32944, 0.30079, and 0.27726. Native excess-RMSE advantage over mean cross-cell medoids was 0.00212 and not established by the exact sign test ($p = 0.25$; Fig. 3). The result demonstrates held-out compatibility and separates formal similarity from empirical response similarity; it does not demonstrate cell-line predictive discrimination.

4.3. Curated Petri nets and interface dependence

Five plant–animal pairs model DNA repair/genome instability (GIM), energy metabolism (DCE), cell-cycle control (SPS), cell death/autophagy (RCD), and immune control (AID). Comparative studies and cancer-hallmark literature guided component selection; dedicated reviews informed plant DNA-damage, cell-cycle, and autophagy mechanisms.^{18–21} Transition-level provenance is machine-checked. These manually curated qualitative nets contain no kinetics, concentrations, tissue context, or independent experimental validation.



(a) *Animal/human GIM model*



(b) *Arabidopsis* GIM model

Fig. 4. Curated GIM Petri nets: (a) animal/human and (b) Arabidopsis. Circles are places, rounded boxes are transitions, brackets identify observables, and gray τ transitions are internal. The diagrams and analysis use the same executable definitions.

Table 4. Curated model-pair outputs. Relations and d_6 are interface-conditional; PN-GDDA is structural.

Module	Process	Formal class	d_6	PN-592	Orbit-576
GIM	DNA repair	weak bisimulation	0.000	0.9976	0.9975
DCE	energy metabolism	weak bisimulation	0.000	0.9969	0.9968
SPS	cell-cycle control	weak bisimulation	0.000	1.0000	1.0000
RCD	death/autophagy	plant in animal	0.143	1.0000	1.0000
AID	immune control	non-comparable	0.429	0.9907	0.9904

Figure 4 exposes the executable GIM encodings. Both include damage sensing, checkpoint control, repair, and restoration. The animal model uses apoptosis, whereas the plant model uses an internal SMR step followed by differentiation or endoreduplication. Their shared `exit_cycle` label is a deliberately coarse observation, not evidence that terminal mechanisms are biologically identical.

The ten nets produced 5–14 reachable states. GIM, DCE, and SPS were weakly but not strongly bisimilar; RCD gave plant-in-animal one-way simulation; AID was non-comparable. Table 4 juxtaposes these classes with native PN-GDDA. The 592- and 576-orbit variants changed no score by 0.001 and no threshold decision: all five pairs were structurally similar at 0.9 despite three different behavioral classes.

The GIM reachable systems make the weak relation inspectable in the repository: damage, checkpoint, repair, restoration, and cycle-exit paths match, while additional internal paths are absorbed by τ closure. A finer interface that distinguishes apoptosis, differentiation, and endoreduplication would alter the classification.

RCD supplies the contrasting directional result: both nets contain stress, mTOR inhibition, autophagy, and survival, but only the animal encoding exposes a canon-

ical caspase-apoptosis branch; the plant branch uses an internal metacaspase event. Silent subdivision preserved every class over 300 refinements per module, while targeted relabeling changed the expected relations. Distances against 2,000 label permutations assess sensitivity inside the curated models, not external biological significance. Full RCD diagrams and robustness plots remain in the reproducible repository outputs.

5. Discussion

5.1. *What agreement means*

The classifier turns an ordered pair of executable models into one statement at a declared observational interface. Proposition 2 supports the reporting priority: strong bisimilarity entails weak bisimilarity, and weak bisimilarity entails both simulation directions. One-way classes retain their orientation and therefore express behavioral containment, not equivalence. The fixed-point theorem concerns the algorithm actually executed, including its in-place deletion order and treatment of τ ; software tests complement that proof but cannot replace it.

The controlled cases also show why structural, linear-time, and branching comparisons answer different questions. Identical unlabeled structure cannot determine labeled behavior (Proposition 4), while Example 1 has identical exact finite traces but different branching. PN-GDDA scored 2/6 on the predeclared behavioral task, although the independent Holmes implementation reproduced every structural score to 12 decimal places. This is not a defect in PN-GDDA: labels, initial states, and reachable branching are outside that statistic by design. LTS-GDA recovered labeled local motifs and scored 4/6, but it still lacked silent abstraction and recursive successor matching. Its strongest held-out association ($\rho = 0.669$) remained inconclusive ($p = 0.097$). The evidence therefore supports complementary reporting of structural and behavioral measures, not replacement of one by the other.

5.2. *Interface and update semantics*

Every relation is indexed by Σ . Declaring an event internal changes the paths hidden by abstraction, while merging outcomes under one label coarsens the question. In the GIM comparison, apoptosis and differentiation or endoreduplication share `exit_cycle`; weak bisimilarity consequently means agreement only at that resolution, not identity of the terminal biological mechanisms. Proposition 3 covers the controlled transition subdivisions used here, but not arbitrary refinements that introduce conflicts, interleavings, or observable choices.

Update semantics is likewise part of the comparison contract. Switching between asynchronous and synchronous execution changed public GINsim behavior and two HPN classes. General-asynchronous, most-permissive, and stochastic conventions were not evaluated.¹⁷ A reported relation should therefore always name both its observable interface and execution semantics.

5.3. Evidence and biological limits

Python and mCRL2 agreed on all 42 strong/weak decisions across the synthetic, public-model, and HPN experiments. The independent attacker–defender checker also agreed on all 67,600 ordered pairs in the complete declared universe of 260 one- and two-state LTSs over $\{a, \tau\}$. These results are evidence against small-system implementation defects, not a proof beyond the theorem. The GINsim transformations test ingestion and execution but do not constitute independent biological labels; the distinct HPN model families and held-out measurements add an empirical check.

That check remains deliberately modest. The HPN analysis contains nine pair-condition observations: the formal-class association with experimental distance was $\rho = 0.091$ ($p = 0.111$), truncated trace distance was also unsupported, and the CASPOTS native-versus-cross advantage was not established ($p = 0.25$). CASPOTS RMSE measures minimum-distance compatibility over admissible trajectories, not a unique time-series prediction.¹² These data do not support prognostic or cell-line-specific claims.

The curated Petri nets omit kinetics, quantities, noise, tissue context, and independent experimental inference. Provenance makes modeling choices auditable but does not turn them into independent evidence. Label-permutation results measure sensitivity inside those constructions, and cross-organism weak relations do not imply organism-level functional equivalence.

5.4. Computational scope and use

The implementation explicitly enumerates states and may store $|S_1||S_2|$ candidate pairs. The scaling experiment reached 145 states and 18,705 pairs, so it does not establish genome-scale performance. Symbolic state representations, partition refinement, and compositional checking are natural extensions. PN-GDDA was applied to native curated nets and canonical state-machine encodings; using it on SBML-qual or Boolean rules would require a non-unique reverse translation. Larger repository models, prospectively expert-labeled relations, biological replicates, blinded interface selection, and additional update semantics are required before broader predictive claims. At present, the framework is best used to audit two explicit qualitative models at a predeclared observational resolution.

6. Conclusion

Comparing executable qualitative biological models requires an explicit account of observables, silent events, initial states, branching, and update semantics. The proposed classifier reports the strongest supported relation from strong bisimilarity through one-way weak simulation, with fixed-point correctness and termination established for finite LTSs. Controlled cases, mCRL2, and an independent simulation game verify distinct parts of the implementation, while the direct PN-GDDA comparison identifies the information unavailable to a purely structural score. GINsim,

HPN-DREAM/CASPOTS, and curated Petri-net applications show that the audit can cross model formats without hiding interface-dependent assumptions. The present evidence supports reproducible model comparison, not prognosis, cell-line specificity, or organism-level equivalence.

Data and code availability

Code, curated definitions, hash-verified public models, normalized response files, metadata, and generated results are available under the MIT License at <https://github.com/cero1979/BisimulationMol>. The repository documents validation, figure regeneration, tests, and notebook execution. No confidential or participant-level data were used.

Declarations

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